OSI vs TCP/IP model: table of layers, protocols and one attack per layer.

OSI vs TCP/IP Model

OSI Model

The Open Systems Interconnection (OSI) model is a conceptual framework that divides network communication into 7 layers.

OSI - 7 Layers

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

TCP/IP Model

The TCP/IP model is a practical networking architecture commonly represented by 4 layers, used for communication across the internet.

TCP/IP - 4 Layers

4. Application
3. Transport
2. Internet
1. Network Access

OSI Layers, Protocols and Common Attacks

OSI layer	Function / protocols	Example attack

7. Application	HTTP, HTTPS, DNS, SMTP	SQL Injection
6. Presentation	TLS, data encoding, encryption	SSL/TLS downgrade
5. Session	Session establishment and management	Session hijacking
4. Transport	TCP, UDP	TCP SYN flood
3. Network	IPv4, IPv6, ICMP	IP spoofing
2. Data Link	Ethernet, ARP, MAC addressing	ARP spoofing
1. Physical	Cables, radio signals, fiber optics	Cable tapping

Note: OSI layers 5 and 6 describe functions rather than universally distinct protocols. TLS can span conceptual layer boundaries, and ARP is commonly associated with the link layer.

TCP/IP Layers, Protocols and Attacks

TCP/IP layer	Corresponding OSI layers	Protocols / technologies	Example attack
Application	5, 6, 7	HTTP, DNS, SMTP, TLS	Phishing
Transport	4	TCP, UDP	SYN flood
Internet	3	IP, ICMP	IP spoofing
Network Access	1, 2	Ethernet, Wi-Fi, ARP	ARP spoofing

Key Differences

OSI model	TCP/IP model
Has 7 layers	Has 4 layers
Conceptual reference model	Practical protocol architecture
Developed by ISO	Developed through DARPA research and standardized through RFCs
Separates session and presentation layers	Includes these functions within the application layer
Mainly used for learning and troubleshooting	Used for real-world internet communication